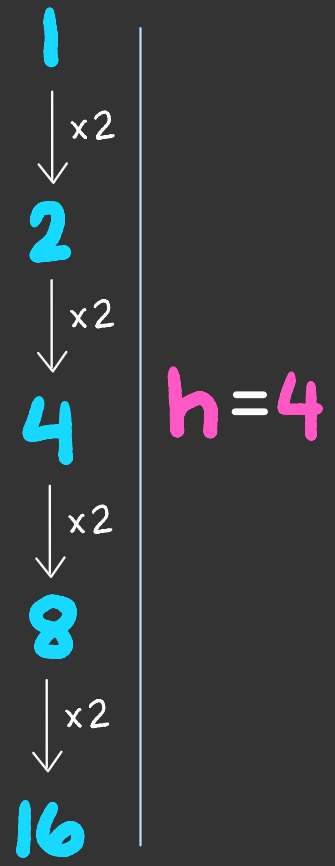
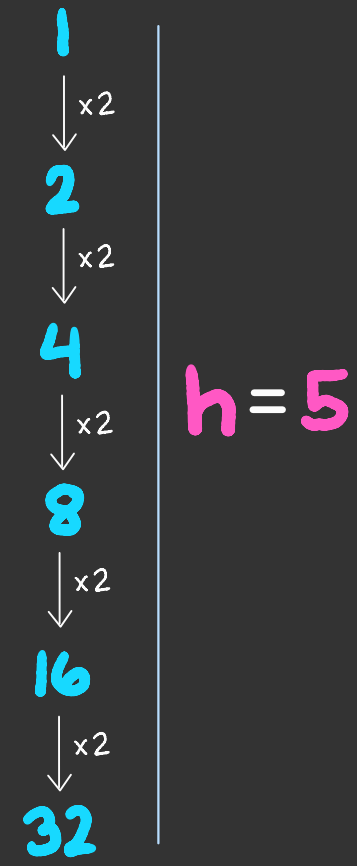
logarithmic complexity

CSCI 134

if we start with the number 1 and double it h times, what do we get?

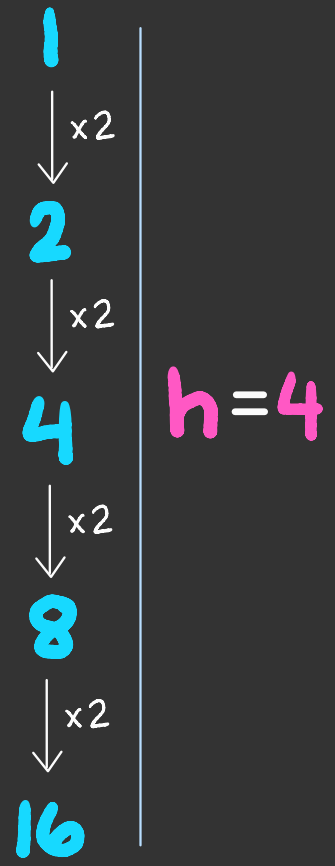
?

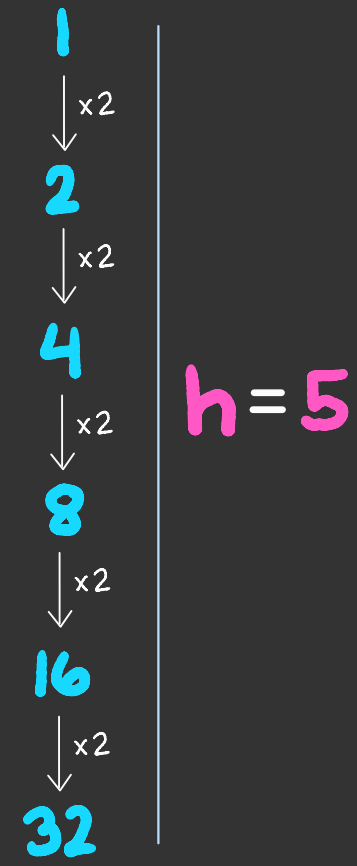




if we start with the number 1 and double it h times, what do we get?

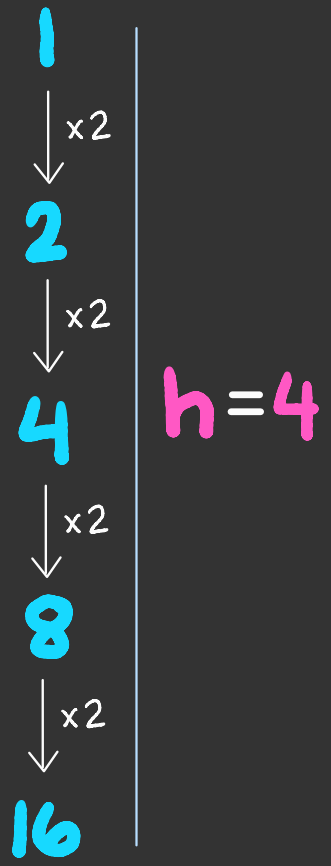
$$2^h$$

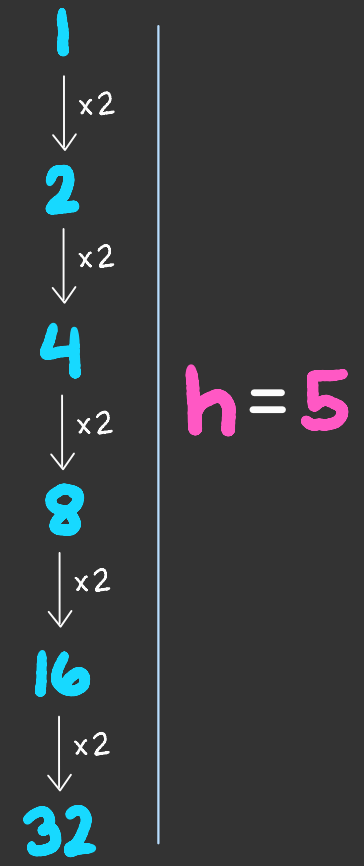




if we start with the number 1, how many times do we need to double it to reach n?

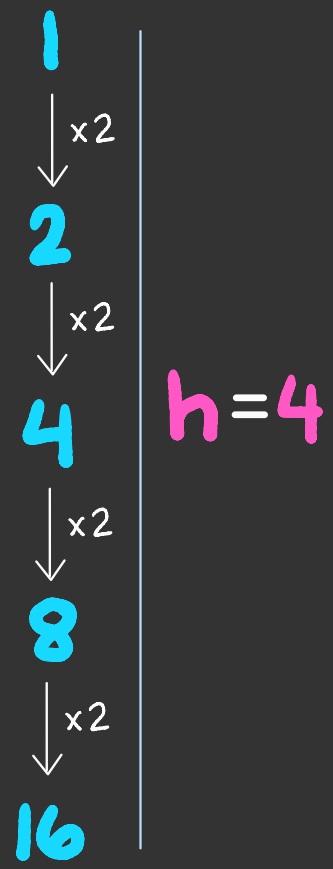
?

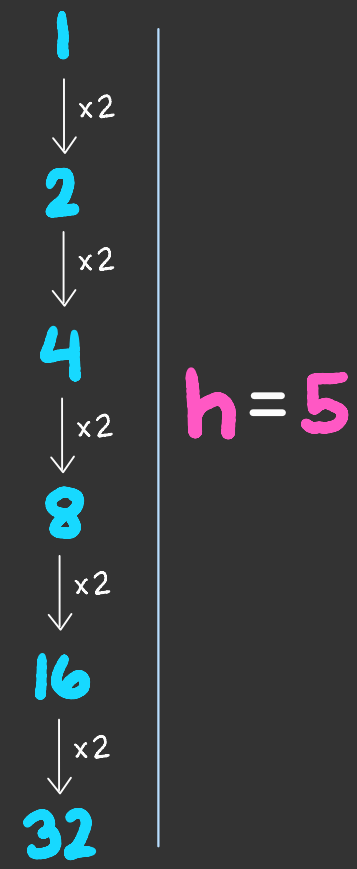




if we start with the number 1, how many times do we need to double it to reach n?

$$\log_2 n$$

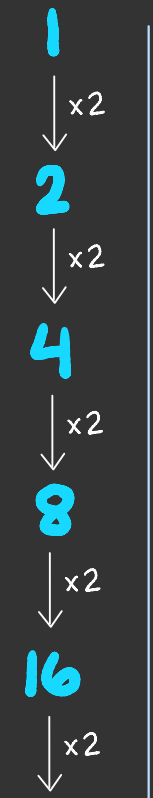




log

is the opposite of

pow



$\log_2 32 = 5$

$pow(2,5) = 32$

log

is the opposite of

pow

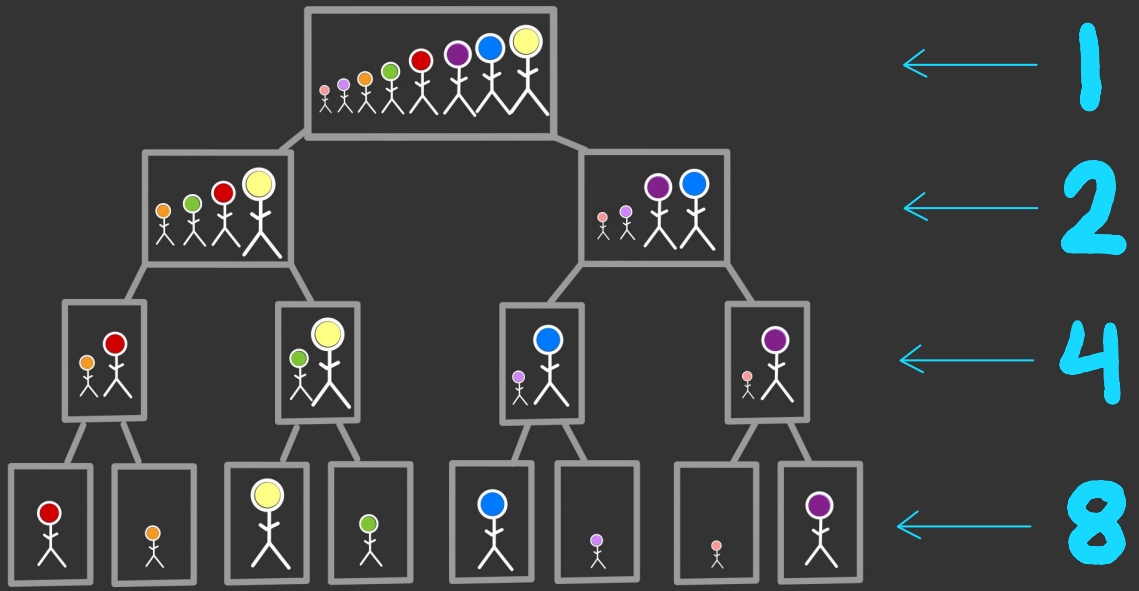
if:

$$2^h = n$$

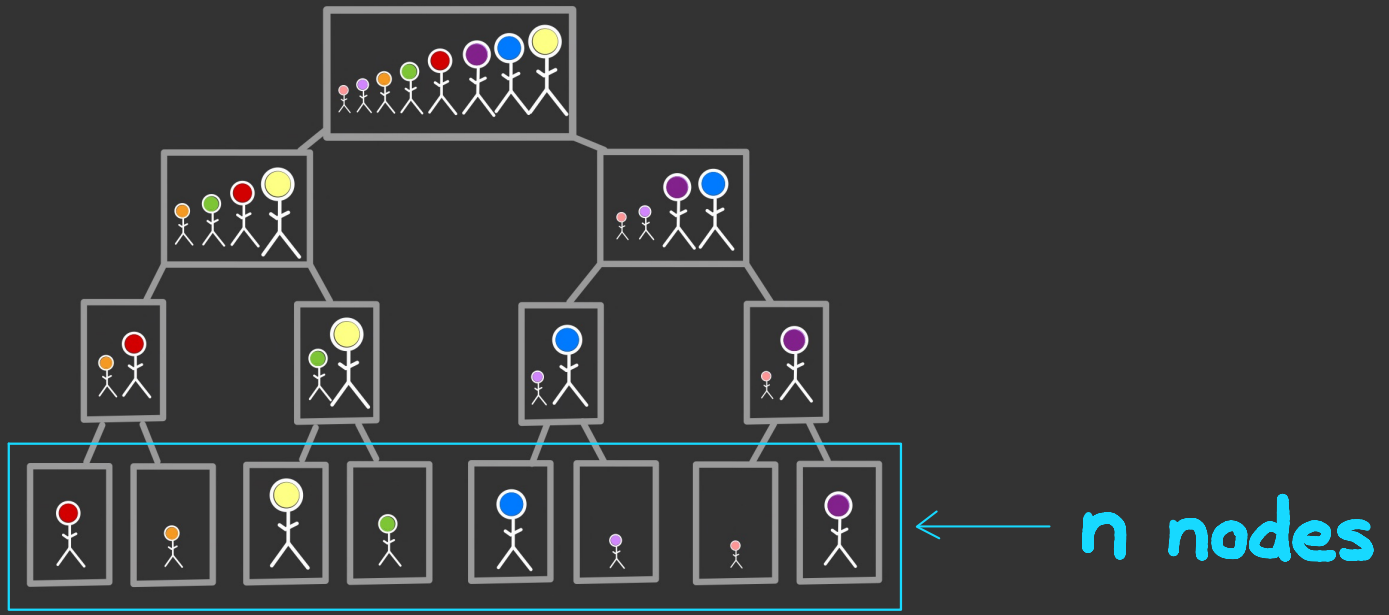
then:

$$\log_2 n = h$$

at each level of the tree, the number of nodes doubles



the final level has a node for each element in the list we want to sort



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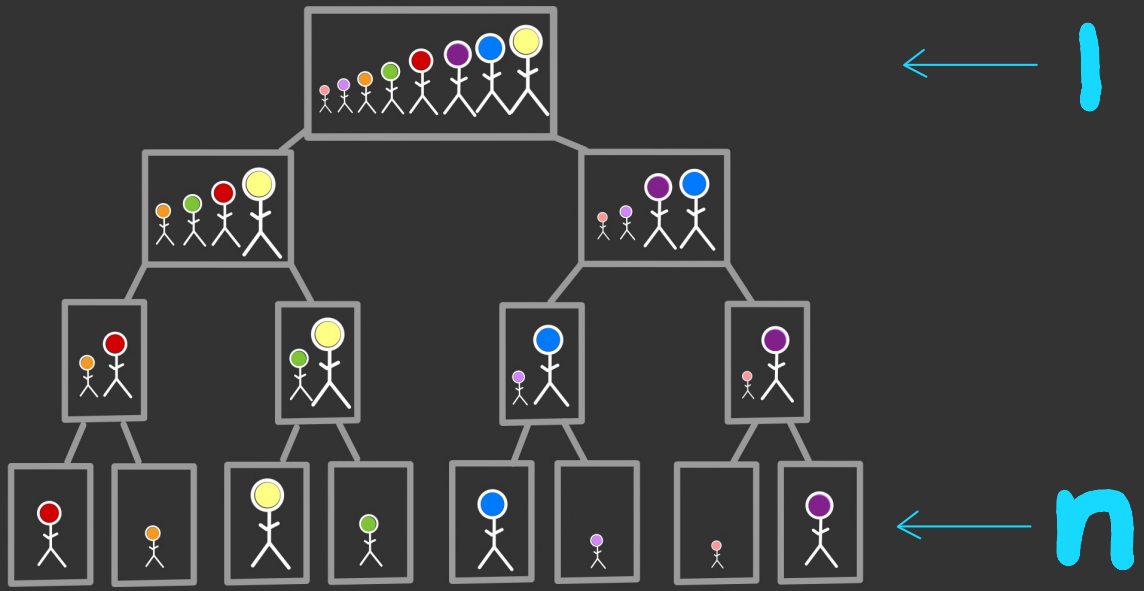

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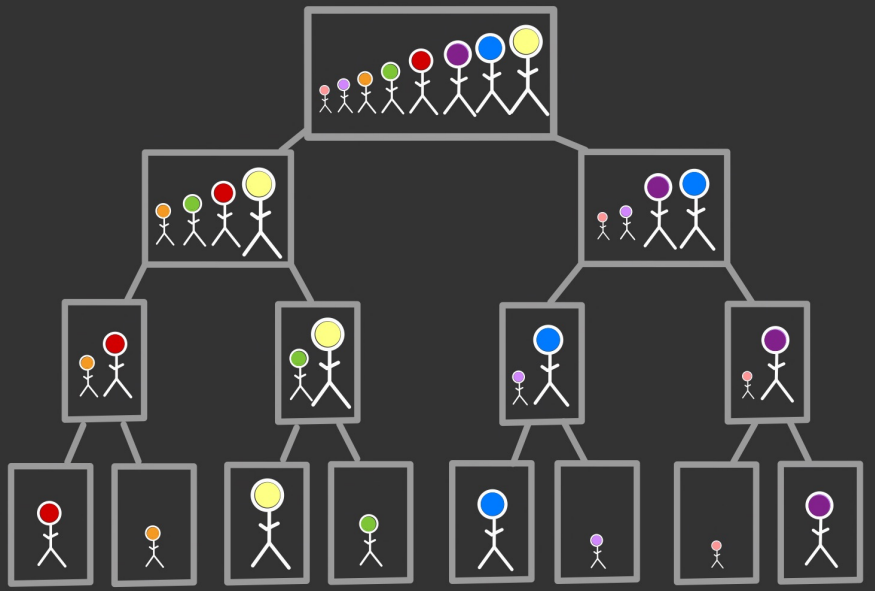
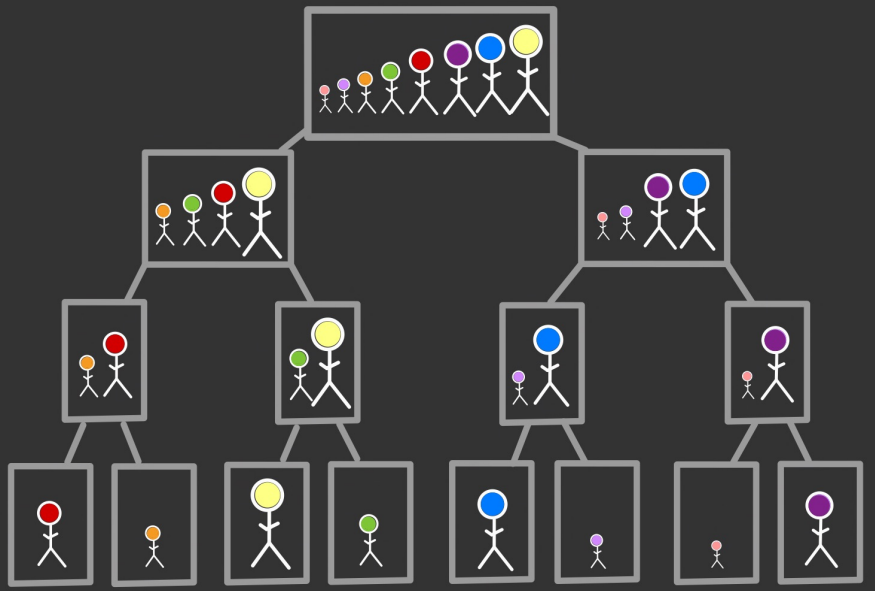


1

$h = \log_2 n$

n

so the height of the tree is $\log_2 n$



1

$h = \log_2 n$

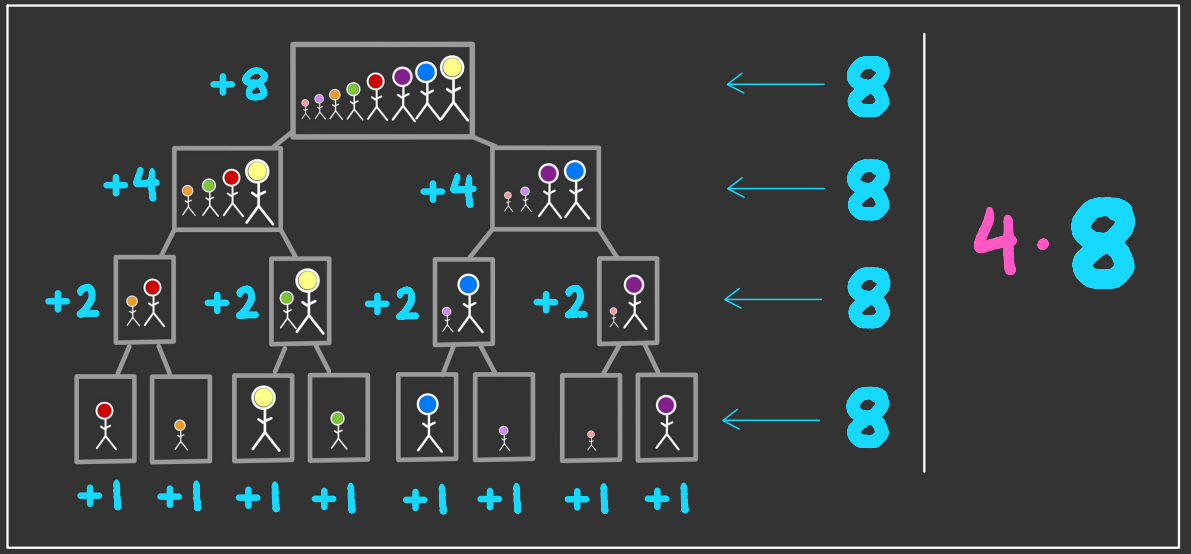
n

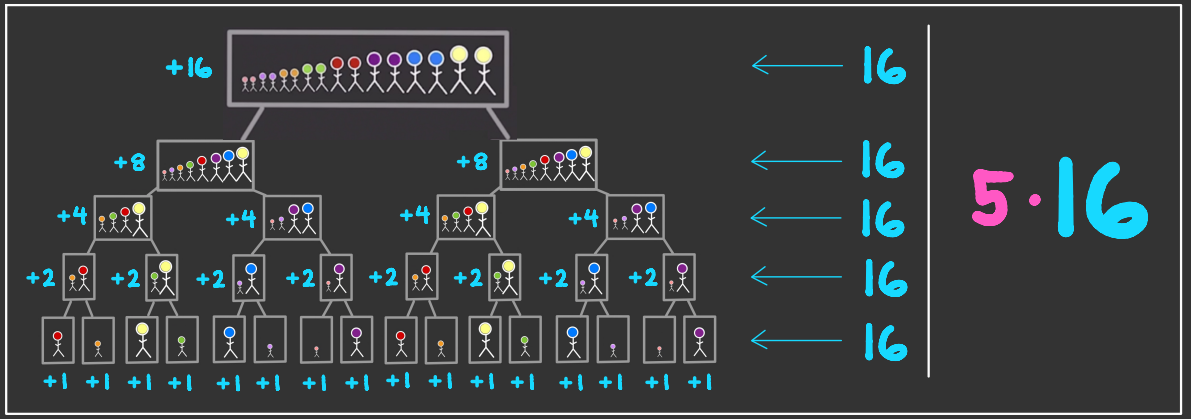
recall:

mergesort makes a total of

hn

comparisons, where h is the height of the tree





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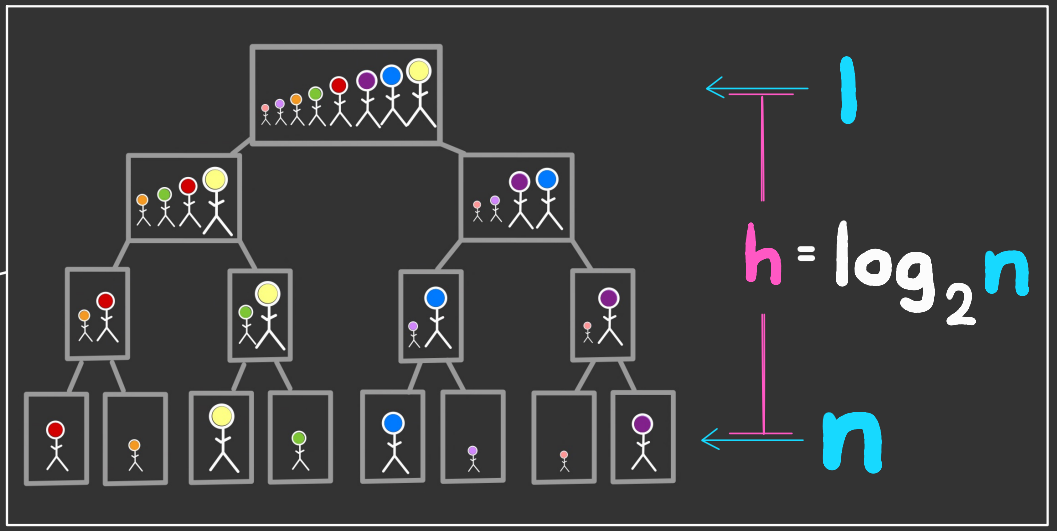
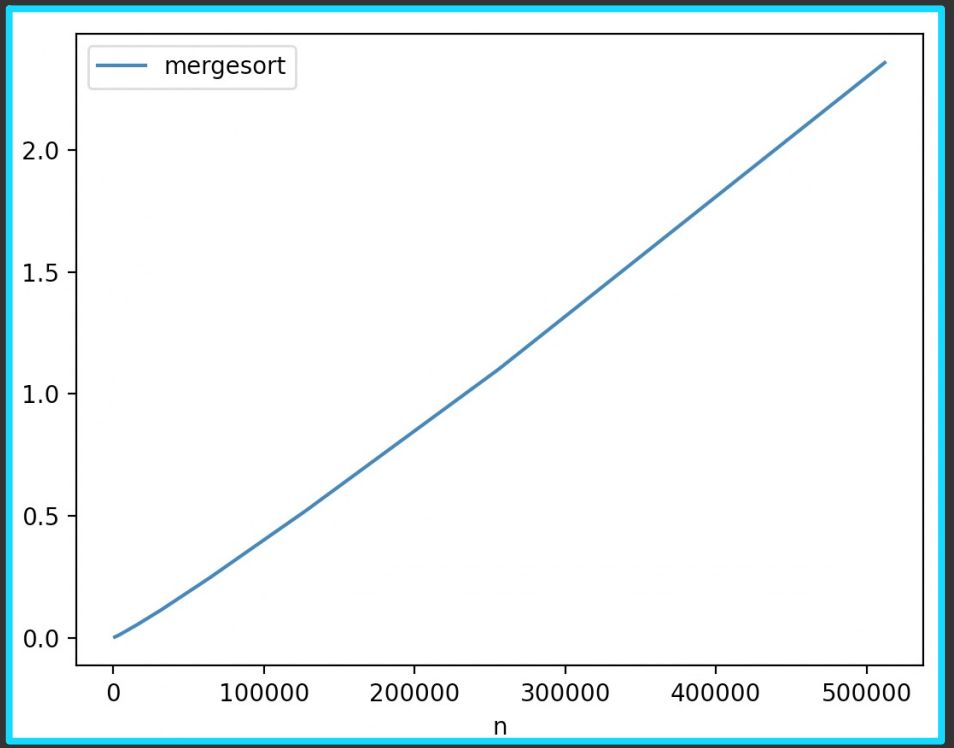
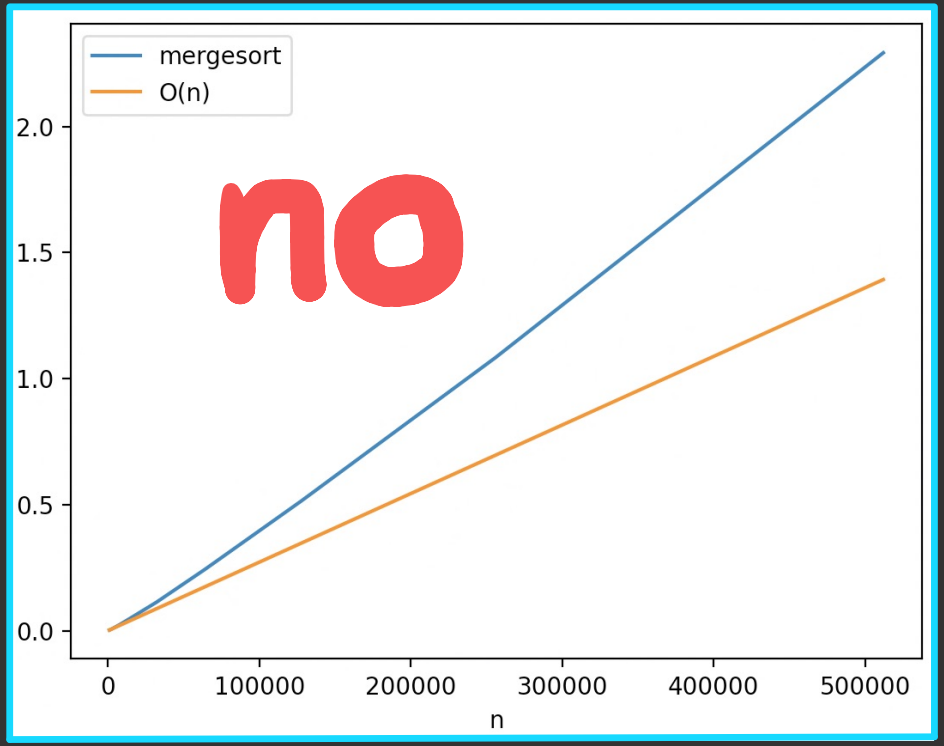


so mergesort makes $n \log_2 n$ comparisons

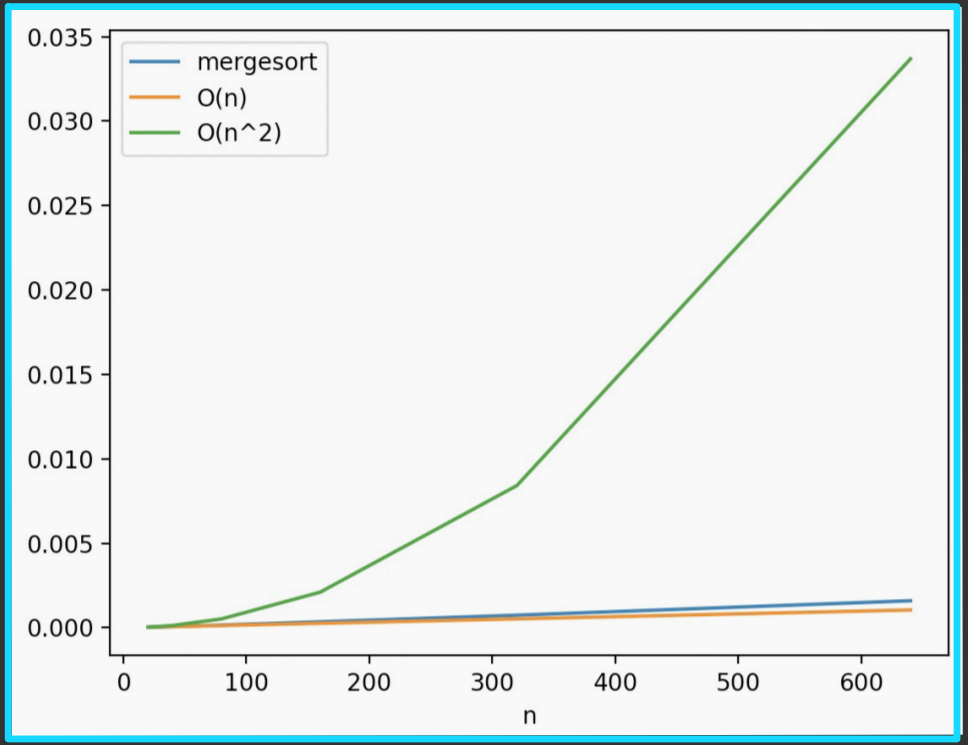
is $n \log_2 n \in O(n)$?



is $n \log_2 n \in O(n)$? ✗



but $n \log_2 n$ is way better than $O(n^2)$



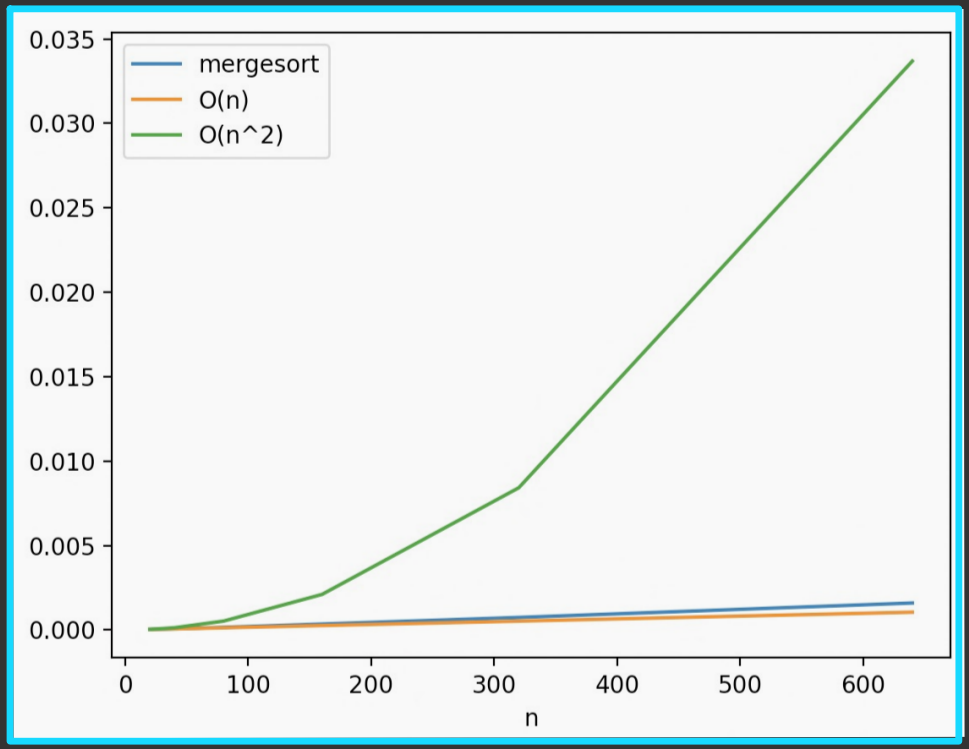
$O(n)$

is better than

$O(n \log_2 n)$

is better than

$O(n^2)$



$O(n)$

is better than

$O(n \log_2 n)$

is better than

$O(n^2)$

turns out this doesn't matter

$n \log_a n \in O(n \log_b n)$

for any positive numbers

a and b

$O(n)$

is better than

$O(n \log_{\square} n)$

is better than

$O(n^2)$

so typically the base is omitted when discussing logarithmic complexity

common complexity classes

slower

$O(1)$	"constant time"	e.g. list indexing
$O(\log n)$	"logarithmic time"	e.g. binary search
$O(n)$	"linear time"	e.g. linear search
$O(n \log n)$	"quasilinear time"	e.g. mergesort
$O(n^2)$	"quadratic time"	e.g. bubble sort